

The whole story in six lines

1. Big projects almost always cost more and take longer than the number the board approved. In oil and gas: 64% over budget, 73% late, 59% average cost growth, and the Middle East is the worst region measured (89% over cost). *(EY, 365 projects over \$1bn)*
2. It isn't a software shortage. It's that the facts that would have saved the project are spread across dozens of systems, dozens of companies and millions of documents nobody can read fast enough.
3. Reading is now cheap. That is the only thing that changed, and it changes which problems are solvable.
4. So we sell three reading machines, in the order the owner can actually buy them.
5. Each one pays for itself out of a single avoided loss on a single project.
6. None of them asks the owner to change how they work. That's the design constraint, not a nice-to-have.

What goes wrong, in money

WHAT HAPPENS	HOW BIG	SOURCE
Contractors ask for roughly twice what they eventually settle for	~2× gap between submitted and settled	IPA via National Academies / Long International
Money in dispute on affected projects	33.4% of the contract budget	HKA CRUX (2,200+ projects)
One average dispute	\$60.1M and about 12.5 months	Arcadis 2025
Redoing work that was built wrong	~5% of project cost (top 10% of projects: 12.4%)	CII IR-153
Where those errors come from	70-80% of cost deviations trace back to design, not building	industry analyses cited in the report
Cost of catching an error late	10× more in construction, 100× more after handover	CII 1-10-100
Overrun after the board has already approved the budget	23% on average, even post-sanction	EY

Plain reading: the money leaks in three places. During the build, through claims. In the design office, through errors that get built. And at the moment of approval, through a number that was too hopeful.

Our three products sit exactly on those three leaks.

Why decades of software didn't fix it

- Every project is a one-off. Teams disband, lessons evaporate, and the next project starts from zero.
- The contract makes truth adversarial. Fixed-price deals reward claiming and reward hiding bad news.
- Low estimates win approval, so estimates come in low. That's not a mistake, it's an incentive.
- Reporting is monthly. Any problem is four to eight weeks old before a leader sees it.
- The decisive evidence is prose: letters, minutes, daily reports, drawing comments. Old software could only handle the tidy 20% of data in databases, and the messy 80% was invisible.

The gap is not analysis, it's reading. That's the sentence the whole business rests on.

The owner's lifecycle, and where the money is really decided

IDEA (screen)	OPTIONS (pick one)	DESIGN (FEED)	APPROVE (the vote)	BUILD (3-6 years)	START-UP (handover)	OPERATE (20-40 yrs)
 cheap to kill	 cheapest moment to change your mind	 where cost is actually set	 the number everything is measured against	 where the overrun becomes visible	 where delay costs most (money spent, no revenue)	 where bad handover data costs for decades

Two facts that drive our whole product order:

1. Cost is **decided** in design and at the approval vote. It only **shows up** years later, during the build.
2. Almost every AI product in this market works during the build, for the contractor. The owner's seat, the design stage and the approval gate are open ground.

The three products on that map

IDEA	OPTIONS	DESIGN	APPROVE	BUILD	START-UP	OPERATE
		[PRODUCT 2]	[PRODUCT 3]	[PRODUCT 1]		
		Design feasibility	FID red team	Claims defence		
		check every revision	pressure-test	& recovery		
			the number	always on		
		~ ~ later opportunities ~ ~				
concept	option	drawing chase	gate challenge	cost forecast truth	handover	asset data
red-team	compare	(deliverables)	book	schedule truth	complete-	feeds the
				interface gaps	ness	next vote

- **Product 1 — Claims defence and recovery.** Ships first. Runs during the build, on the record the owner already controls.
- **Product 2 — Design feasibility.** Ships alongside or just after. Runs during design, on the owner's own drawings.
- **Product 3 — Approval red team (FID).** Switched on after 1 and 2 have earned trust, because it needs the owner's history and it delivers an unwelcome message at the highest-stakes moment.

The greyed items are the roadmap. Each one reuses the same reading layer, so the second product costs a fraction of the first.

One project, one team, one week: the scenario the rest of this deck uses

The project. Jebel Nasr Train 4, a \$5.2bn gas processing train for a Gulf national oil company. Lump-sum EPC contract with Kaisan Engineering & Construction (KEC), signed with 14 amendments to date. Month 26 of a 48-month build. Late-delivery damages run at \$45k per day, capped at 10% of contract value. Claim notices are due within 28 days of the event, and must go to the named recipient.

The people.

NAME	ROLE	WHICH PRODUCT
Faisal Al-Harbi	Owner's contracts and commercial manager, Jebel Nasr Train 4. Team of four.	Product 1, the buyer and the user
Mariam Kassab	Senior quantity surveyor in Faisal's team	Product 1, second user
Omar Zahra	Owner's in-house counsel	Product 1, review routing
Priya Raghavan	Owner's design and engineering manager	Product 2, the user
Tom Beckett	Project director, Jebel Nasr	Product 3, the sponsor
Layla Haddad	Lead estimator, capital projects	Product 3, the user who gets challenged

Day one, the only setup Faisal does: five read-only connections. No migration, no new system of record.

SOURCE	WHAT IT IS	VOLUME AT CONNECTION
\JN4\02-Contract\ on SharePoint	The EPC contract, 14 amendments, the particular conditions	61 PDFs
JN4-correspondence-2024-2026.zip exported from Aconex	Letters, transmittals, notices, minutes	11,482 documents
\JN4\07-Site\DSR\	Daily site reports, about 30 a week	3,140 PDFs
JN4-P6-2026-07.xer	The monthly schedule update from P6	1 file, 8,900 activities
KEC-IPA-039.pdf	The current interim payment application	1 file

Names, clauses and companies are illustrative for a case study.

Product 1: what it actually does, in Faisal's week

For Faisal Al-Harbi, the owner's contracts and commercial manager on Jebel Nasr Train 4. His team of four connects the existing record once, then starts with the event that matters: on 3 August 2026, KEC submits claim 07, KEC-CL-007 .pdf, for \$40.2M and 96 days of extension, built on six delay events.

The agent reads the contract, letters, schedule update and site reports overnight, then gives Faisal a cited position:

- **Event 2 notice was late.** KEC-L-1987, dated 41 days after the event in DSR-2026-02-14. The 28-day requirement means 13 days late, so the event is barred.
- **Event 4 has no notice at all.** No notice appears in the 11,482 documents.
- **Event 5 went to the wrong recipient.** KEC-L-2043 is addressed to the site manager, not the named recipient. Defective notice.
- **Events 1, 3 and 6 were properly notified.** Their notices are found and valid.
- **Position.** 41 documents cited. \$40.2M claimed, less \$12.4M barred, less \$5.6M of quantum the record does not support, leaves **\$18.0M defensible**.
- **Milestone 4 was certified 47 days late.** Certificate JN4-MC-004 against the milestone date in the contract means 47 days at \$45k, or **\$2.1M** chargeable this month.
- **An unregistered exposure.** KEC-L-2214 of 14 May 2026 signals a claim not in the risk register: **\$3.3M**, with 6 days left to reply.

Every item shows the money at stake, how confident the agent is, and the documents behind it. Faisal approves, edits or rejects. The agent writes the letter. A person sends it.

Two things make it credible: it never sends anything itself, and every line clicks through to a page in a real document.

How Product 1 works, end to end



Autonomy, stated plainly: the agent is autonomous about reading, linking, calculating and proposing. It has no autonomy over anything that leaves the building.

Product 1: what it's worth, and what it costs you

Value (money). On a \$5bn project, our conservative model:

STEP	NUMBER	WHERE IT COMES FROM
Claims submitted over the build	~\$400M (8% of project value)	our assumption, to validate with the owner's own history
Typically settled at about half	~\$200M paid	IPA: submitted vs settled gap "as much as half"
Sharper, evidenced defence on 10% of that	\$20M kept	our model
Late-delivery damages and back-charges pursued properly (from ~40% to ~70% of what's owed)	~\$9M recovered	our model, using a \$30M entitlement pool
One dispute kept out of formal proceedings	\$5–8M of legal and expert cost avoided, plus 12.5 months of management time	Arcadis: \$60.1M and 12.5 months per average dispute
Total per project	~\$30M	
Price at 0.05% of project value	\$2.5M	industry pricing runs 0.1–0.3% of project value
Return	~10:1	

Likelihood of delivering: high. The evidence lives in documents the owner already holds. Deadline maths is arithmetic, not judgement. Every output is checkable against a cited page. The category is already proven commercially: Document Crunch reached 400+ customers and Trimble bought it for \$246.4M in April 2026.

Effort from the persona: low. Give us a read-only copy of the contract and the correspondence archive. Spend about 20 minutes a day on a ranked queue. Nothing to migrate, no new system of record, no change to the monthly cycle.

Sacrifice from the persona: real but answerable. They're letting software read the most commercially sensitive, sometimes legally privileged, material on the project. Our answer: in-Kingdom or in-UAE hosting, private cloud, no training on their data, access mirroring the existing

Product 2: what it actually does, in Priya's week

For Priya Raghavan, the owner's design and engineering manager on Jebel Nasr Train 4. On 12 June 2026, the design contractor issues sheet A-207 rev C for the process area. Priya does nothing different: the revision lands in the same transmittal folder it always did.

Three checks run in four minutes and hand back a cited pass or fail list:

- **Legal.** Fire separation is 1.2 m short. The fire chapter of the Saudi Building Code, clause 7.4.2, is quoted verbatim with the page, next to the drawing region.
- **Affordable.** The change costs \$6.8M against \$4.1M of remaining design allowance. The cost model line and the approved budget ceiling are shown.
- **Buildable.** Moving the 132 kV transformer adds 14 weeks to a long-lead order already on the critical path. The purchase order date and the P6 activity are shown.

Priya accepts the first finding, disputes the second as "already waived by the authority", and issues the revision with the exception recorded. The dispute becomes a labelled error we report on.

It's a spell-check for feasibility. It catches the error while it costs thousands, not the tens of millions it becomes once it's in the ground.

Strongest demand signal in the market: Dubai Municipality is building a system to issue permits automatically by reading submitted drawings against the Dubai Building Code. When the regulator automates the check, every owner has to arrive pre-compliant.

How Product 2 works, end to end

INPUT	PROCESSING	ACTION	DETERMINISTIC OR AGENTIC
code edition, pinned by Priya on day one budget ceiling, pinned	→ index clauses, keep the page and the exact text	[code]	
	→ store as a number	[code]	
drawing / document revision (A-207 rev C)	↓		
	→ detect the revision, diff against rev B	[code]	→ start a check run
	read geometry, labels, specified materials	[model]	deterministic trigger (no human asked for it)
	↓		
	LEGAL clause retrieval	[code]	
	clause-to-drawing comparison	[model]	→ finding, clause quoted verbatim + drawing region
	AFFORDABLE cost delta	[code]	→ \$6.8M vs \$4.1M left
	BUILDABLE clash geometry	[code]	deterministic
	long-lead dates	[code]	→ 14 weeks on a critical path order
	deterministic		
	↓		
	citation check	[code]	→ no clause text, no finding
	deterministic veto		
	↓		
	HUMAN VERDICT	[human]	→ accept / dispute / accept with a waiver
	advisory, never blocking: the revision can always be issued		
	↓		
	record the dispute reason	[code]	→ a labelled error
	deterministic learning signal		

Product 2: what it's worth, and what it costs you

Value (money). On a \$5bn project:

STEP	NUMBER	SOURCE
Rework at ~5% of project cost	~\$250M	CII IR-153
Share caused by design (~28%)	~\$70M	industry analyses in the report
Catch half of it in design	~\$35M avoided per project	our model on sourced inputs
On a \$20bn project, the design-caused pool	~\$280M	same method
Review time	6 weeks down to 3 days on Saudi Building Code review	WhiteHelmet customer testimonial, 2026 (self-reported)
Independent return modelling	4:1 to 30:1 cost avoidance; a query costs \$200–500 at drawing stage vs \$2,000–8,000 in the field	Mirage Metrics, 2026

Likelihood of delivering: high on a narrow scope, medium if we promise everything. Reading drawings and code is now doable, and two Gulf vendors already sell a version of it. The honest risk is coverage: we start with one code set and the disciplines that cause the most rework, and we say so.

Effort from the persona: lowest of the three. It hooks into the design file drop that already happens. One-time job: confirm which code edition and which budget ceilings apply. After that the designer's habit doesn't change at all.

Sacrifice from the persona: professional pride, mostly. A machine is marking an engineer's work inside the engineer's own workflow. Our

Product 3: what it does, and why it waits

For Tom Beckett, project director of Jebel Nasr, and Layla Haddad, lead estimator for capital projects. Product 3 runs once per gate, later. Train 5 goes to the investment committee with a \$6.1bn estimate and 8% contingency.

Before it is used forward, the agent runs backwards on 14 of the owner's own finished projects and shows what it would have said about each. Layla gets the challenge book first, privately. The agent asks nine questions and casts no vote.

The agent pressure-tests the number: it compares this project against how comparable past projects actually turned out, audits the design package for the gaps that predict overruns, and writes the challenge book for the gate meeting.

Why it waits: it needs the owner's own project history, and it tells the most senior people in the room that their number is wrong. That's a sale you make once you're already trusted and already inside the data. Products 1 and 2 buy that position.

The prize is the biggest, though: EY finds a 23% average overrun **after** approval. On a \$5bn approval, that's \$1.15bn of exposure. Taking a fifth of it off the table is \$230M.

How Product 3 works, end to end

INPUT	PROCESSING	ACTION	DETERMINISTIC OR AGENTIC
14 finished owner projects	→ extract sanctioned vs final cost and date	[model] [code]	
	↓		
	REFERENCE CLASS: median growth, spread	[code]	
	↓		
this gate's estimate, basis of estimate, schedule, contingency	→ definition completeness audit against the owner's own gate standard contingency vs the reference class	[model] → what is missing from the definition at this gate [code] → 8% asked, 23% median historical growth	agentic proposal deterministic
	↓		
	citation check	[code] → every number traced to a source document	deterministic veto
	↓		
	PRIVATE PRE-READ to the estimator and sponsor challenge book: nine questions with evidence	[human] → the estimator sees it first and can respond [model] → questions, not a recommendation	no surprise in the room the agent never votes

The agent never votes. It gives Layla and Tom a cited set of questions before the gate meeting.

The three side by side

	1. CLAIMS DEFENCE & RECOVERY	2. DESIGN FEASIBILITY	3. APPROVAL RED TEAM
Persona	Contracts / commercial manager	Design / engineering manager	Project director, estimator, investment committee
Stage	Build	Design	The approval vote
Value per \$5bn project	~\$30M	~\$35M	~\$230M of exposure addressed
Likelihood of delivering	High	High on a narrow scope	Medium
Effort from the persona	Low: read-only files, 20 min/day	Lowest: hooks existing file drop	High: hand over portfolio history
Sacrifice	Trusting software with privileged commercial material	Being marked by a machine	Political: an unwelcome message at the biggest vote
Data needed	Contract, letters, schedule, site reports	Drawings, code set, budget ceilings	Historical outcomes across the portfolio
Verdict	Ship first	Ship in parallel	Earn the right, then ship

The rule behind the order: value divided by (effort + sacrifice), with delivery confidence as the multiplier. Product 3 wins on value alone and loses on everything else, which is exactly why it's third and not never.

The adoption promise (this is the real product decision)

The fastest way to lose this market is to ask a project team to change how it works mid-project. So we commit to six rules:

1. **Runs on files they already have.** Read-only. No migration, no new system of record.
2. **No new habit.** It arrives where they already look (email or Teams), and reviews happen on one screen.
3. **The agent never sends.** It drafts. A person presses send. Always.
4. **Every claim is clickable.** No assertion without a citation to a page in a real document.
5. **Proof on history first.** We run on last year's closed files and show what it would have caught, before it touches a live project.
6. **Priced against one avoided loss,** and sized so a pilot sits inside one project team's own authority.

That last one matters commercially: ADNOC signed a \$340M three-year agentic-AI contract, but only after a proof of concept. The deal shape in this market is small paid pilot, then large multi-year contract.

Proof from people who live it

"The difference between the amount submitted for a claim versus the amount actually paid at settlement... is as much as half. The owner too must ensure profitability of his project by defending unsupportable claims by a 'low ball' contractor." IPA via National Academies and Long International. *Product 1, stated from the owner's seat.*

"What most projects lose to poor claims management is not a dispute they should have won, but entitlement they never assembled the evidence to claim." Kairos, claims advisory, 2026. *The failure mode named exactly.*

"A fully integrated digital system capable of issuing building permits automatically and without human intervention by reading submitted drawings and documents, verifying compliance with the Dubai Building Code." Dubai Municipality, 2026. *Product 2's demand signal, from the regulator.*

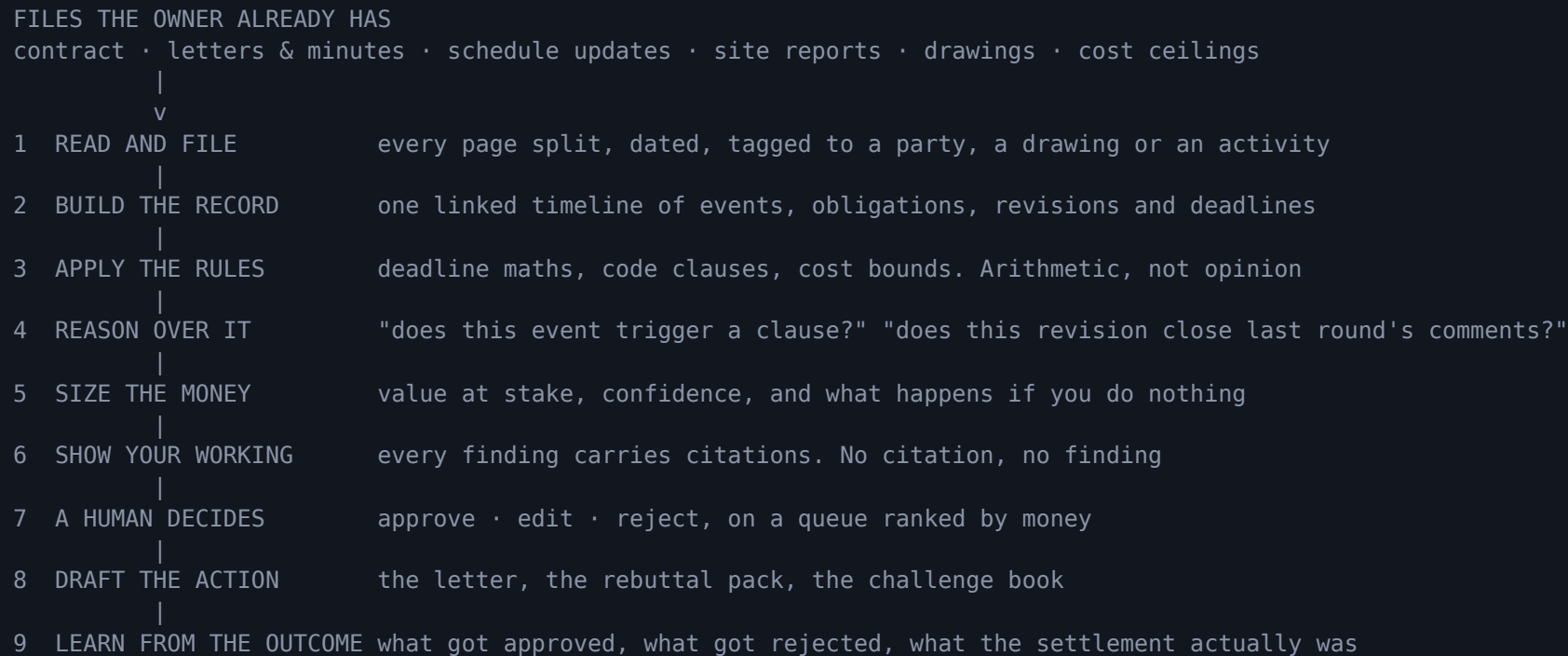
"We reduced our compliance review cycle from 6 weeks to 3 days. The AI caught two fire environmental gaps that would have delayed our building permit by months." WhiteHelmet customer, 2026 (self-reported). *Product 2, quantified in the Gulf.*

"More than half of its potential lost to underperformance... all shared a root cause: bias." McKinsey on an owner's capital portfolio, 2025. *Product 3's reason to exist.*

"ADNOC is on a mission to become the world's most AI-enabled energy company." \$340M, three years, after a proof of concept. Musabbeh Al Kaabi, ADNOC Upstream CEO, 2025. *The willingness to pay, and the price ceiling.*

Honest gap: no on-record Aramco or ADNOC executive naming a claims, estimating or design-check tool specifically. Owner demand for Product 2 is inferred from the regulator's move and the mandatory building-code reset. We validate that in primary interviews.

How it works, in one picture



Two engineering commitments behind that picture:

- **The hard maths is deterministic, the soft reading is the model's job.** Deadlines, damages and cost bounds are computed in code so they're always reproducible. The model reads and proposes, it doesn't do the arithmetic that ends up in a letter.
- **Nothing ships without an eval.** Each skill has a scored test set built from closed projects: did it find the events a specialist found, did it invent anything, did it get the deadline right. Releases are gated on those scores.

How we'll know it's working

Business. Money kept or recovered per project (the number the pilot is judged on), pilot to contract conversion, share of project value captured in price.

Product. Items reviewed per week per manager, share of drafted actions actually sent, time from an event appearing in the record to a human deciding on it, and one honest counter-metric: how often a manager overrides the agent, and whether that falls.

AI engineering. Recall against a specialist's own findings on closed projects, invented-fact rate at zero tolerance for cited claims, deadline accuracy, and cost per document read.

What success looks like for the agent

Four gates, in order. Nothing moves to the next gate until the previous one passes. Each gate has a pass mark and a stop rule, and each is measured on the owner's own closed projects before it is measured on a live one.

Gate 0. Reading is trustworthy (offline, closed project archive)

WHAT WE MEASURE	PASS MARK	WHY THIS NUMBER
Documents read and correctly typed	98% or better	Below this, the record has holes and every later number is suspect
Dates extracted correctly	99.5% or better	A wrong date is a wrong deadline, which is the one error the product cannot survive
Documents refused and queued for human	Reported, not hidden; under 5% of volume	Abstaining is allowed. Guessing is not.
Invented facts	Zero. Any occurrence blocks the gate	This is a legal record.

Gate 1. Findings are worth a person's attention (offline, against a specialist's own conclusions)

WHAT WE MEASURE	PASS MARK
Precision on findings shown as findings	85% or better
Recall against commercial team’s closed-project findings	80% or better, and at least one material item they missed
Evidence chain accuracy	99.5% or better
Deadline arithmetic	100%
Low-confidence items shown as “watching”	100%

Stop rule: precision below 60% after eight weeks means return to Gate 0.

What success looks like for the agent (continued)

Gate 2. A human agrees, in practice (shadow mode on a live project, nothing sent)

WHAT WE MEASURE	PASS MARK
Findings actioned rather than only seen	80% or better
Drafts approved with minor edits or none	70% or better
Override rate	Falling week on week
Digest ignored two days running	Under 10% of weeks
Cost per document read	Under \$0.05 blended

Stop rule: four straight weeks without review pauses the pilot.

Gate 3. It changed the money (live pilot, the number the pilot is judged on)

WHAT WE MEASURE	PASS MARK
Exposures surfaced and countersigned	\$20M or more per pilot project
Notice deadlines missed on covered scope	Zero
Money kept or recovered agreed before pilot	Beats pilot fee by 5x or better
Findings surviving to settlement	Reported honestly, including failures

The one number for each product

- Product 1: money kept or recovered per project, countersigned by the customer.
- Product 2: share of flagged findings that would have become field changes, measured on closed revision history, with dispute rate as honest counter-metric.
- Product 3: gate decisions changed, meaning contingency raised, scope re-cut, or approval deferred, with estimator agreement that the question was fair.

Risks, and what we do about them

RISK	WHAT WE DO
Owner won't give a machine the privileged commercial record	Start on closed projects. In-region private cloud. Permissions that mirror the existing team. Full read audit.
Wrong finding damages a legal position	Draft only, never send. Everything cited. Deadline and damages maths in code, not model output.
Code coverage in Product 2 is thinner than promised	Narrow first release: one code edition, the disciplines that cause the most rework. Publish what's in scope.
Incumbents move (Trimble, Procore, and the AI vendors)	They're contractor-side and build-phase. We're owner-side across design, approval and build, on data they don't have.
Product 3 needs history the owner may not have organised	Bundle the history clean-up as the first paid step, and lean on public outcome data until it exists.
Buying gates in the Gulf (registration, local content, cyber certification, data residency)	Treat them as features on the roadmap from day one, not paperwork at the end.

First 90 days

- **Days 1-15.** Ten interviews: owner contracts managers, design managers, one estimator, one investment-committee member. Validate the \$400M claims-submission assumption against a real portfolio.
- **Days 16-45.** Product 1 on one closed project's archive. Success test: it finds what the commercial team found, plus at least one thing they missed, all cited.
- **Days 30-60.** Product 2 on one FEED revision history. Success test: it flags the errors that actually became field changes.
- **Days 60-90.** One paid pilot signed on a live project, with the money-kept metric agreed in writing before we start. Compliance work (hosting, cyber certification, local content) running in parallel.

Product 3 gets a retrospective demo only in this window: "here are the twelve flags we'd have raised at your last approval." No sale attached yet.

Where this goes

Same reading layer, more skills, in the order the owner will accept them:

- **During the build:** is the cost forecast honest, is the schedule telling the truth, are the gaps between contractors going to bite.
- **In design:** which of the 3,900 late documents actually threaten the finish date.
- **At handover:** what's missing before operations will accept the asset.
- **Then the loop closes:** every finished project becomes evidence for the next approval vote. That history is the owner's own, and it compounds, which is the part no competitor can copy.